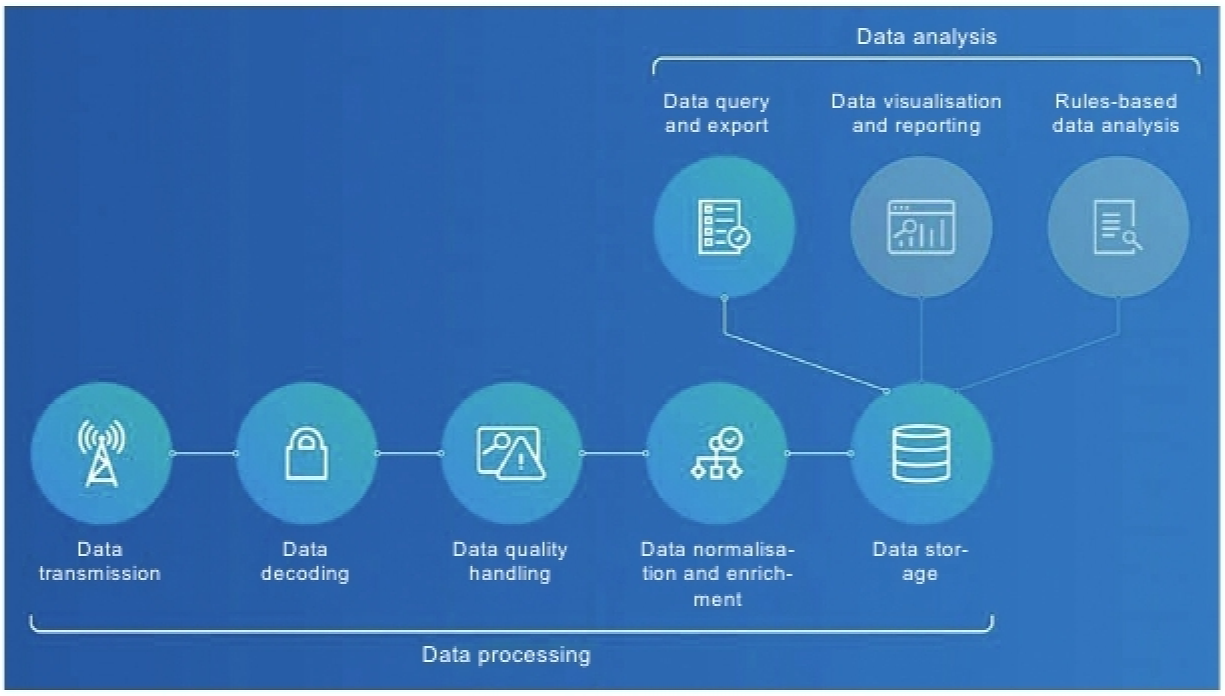
Data Management Systems for IoT Devices

The Internet of Things (IoT) device management enables users to track, monitor and manage the devices to ensure these work properly and securely after deployment.



Steps in IoT data management (Credit: *https://blog.bosch-si.com*)

Billions of sensors interact with people, homes, cities, farms, factories, workplaces, vehicles, wearables and medical devices, and beyond. The Internet of Things (IoT) is changing our lives from managing home appliances to vehicles. Devices can now advise us about what to do, when to do and where to go. Industrial applications of the IoT assist us in managing processes, and predicting faults and disasters. The IoT platforms help set and maintain parameters to refine and store data accordingly.

Data management is the process of taking the overall available data and refining it down to important information. Different devices from different applications send large volumes and varieties of information. Managing all this IoT data means developing and executing architectures, policies, practices and procedures that can meet the full data lifecycle needs.

Things are controlled by smart devices to automate tasks, so we can save our time. Intelligent things can collect, transmit and understand information, but a tool will be required to aggregate data and draw out inferences, trends and patterns.

Developers and manufacturers of embedded systems and devices need to build systems that answer the demands of data management. They need to design a data management framework compatible with all the software and hardware that play a role in collecting, managing and distributing data. The design needs to be efficient to accelerate time-to-market of the end-product.

Data from IoT devices is used for analytical purposes. Information that businesses collect and store but remains relatively stagnant, because it is not used for analytical purposes, is called dark data. It includes customer demographic information, purchase histories and satisfaction levels, or general product data. To better understand customers, dark data is invaluable to businesses, as it allows them to uncover additional insights more efficiently.

Before the release of a product, IoT data management requires field tests. Data from the field tests helps improve the design and create a higher-quality product. Collecting field data post-launch helps in continuous product improvement with software updates and by identifying anomalies. This also provides important insights to support the development process of new products.

The IoT data management

In edge computing, data is processed near the data source or at the edge of the network. While in a typical cloud environment, data processing happens in a centralised data storage location. By processing and using some data locally, the IoT saves storage space for data, processes information faster and meets security challenges.

Edge computing, data governance policies and metadata management help firms deal with issues of scalability and agility, security and usability. This further helps them decide whether to manage data on the edge or only after sending it to the cloud.

Sensors produce a large amount of data for edge gateway devices so that these can make decisions by analysing the data. These high-performance systems not only need to collect data in real-time but also to organise and provide data to other systems.

Sensors and devices can connect indirectly through the cloud, where data is centrally-managed, or send data directly to other devices to locally collect, store and analyse the data, and then share selected findings or information with the cloud. Edge devices for data management help secure the most valuable data and